KAIZEN MATHS

Lesson starter

Inverse Trig Differentiation

Core Derivatives and Evaluation

Name _____

S1

Differentiate with respect to x.

$y = \cos^{-1}(-7x - 9)$

S2

Differentiate with respect to x.

$y = (\sec^{-1}(-5x + 3))^4$

S3

Differentiate with respect to x.

$y = \sec^{-1}(7x + 4), \quad x = -1$

S4

Differentiate with respect to x.

$y = \cot^{-1}(5x - 2)$

Show your working

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Exit ticket

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E1

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Differentiate with respect to x.

$y = \sec^{-1}(5x - 3), \quad x = 1$

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Differentiate with respect to x.

$y = \tan^{-1}(4x + 9)$

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Differentiate with respect to x.

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Differentiate with respect to x.

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Show your working

Teacher answers & working

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S1

Differentiate with respect to x.

$y = \cos^{-1}(-7x - 9)$

Answer:

$$\frac{dy}{dx} = \frac{7}{\sqrt{1 - (-7x - 9)^2}}$$

Working:

Identify the inner function: $u = -7x - 9$.

Differentiate the inner function: $\frac{du}{dx} = -7$.

Select the inverse trig derivative rule: for $y = \cos^{-1}(u)$, $\frac{dy}{du} = -\frac{1}{\sqrt{1 - u^2}}$.

Use the chain rule: $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$.

$$\frac{dy}{dx} = (-\frac{1}{\sqrt{1 - (-7x - 9)^2}})(-7)$$

$$\frac{dy}{dx} = \frac{7}{\sqrt{1 - (-7x - 9)^2}}$$

S2

Differentiate with respect to x.

$y = (\sec^{-1}(-5x + 3))^4$

Answer:

$$\frac{dy}{dx} = -\frac{20}{|-5x + 3|\sqrt{(-5x + 3)^2 - 1}} (\sec^{-1}(-5x + 3))^3$$

Working:

This has an outer power and an inverse trig inner expression.

Let $v = \sec^{-1}(-5x + 3)$, so $y = v^4$ and $\frac{dy}{dv} = 4v^3$.

For $v = \sec^{-1}(u)$, $\frac{dv}{du} = \frac{1}{|u|\sqrt{u^2 - 1}}$.

The linear inner function is $u = -5x + 3$, so $\frac{du}{dx} = -5$.

$$\frac{dy}{dx} = \frac{dy}{dv} \times \frac{dv}{du} \times \frac{du}{dx}$$

$$\frac{dy}{dx} = (4v^3)(\frac{1}{|u|\sqrt{u^2 - 1}})(-5)$$

$$\frac{dy}{dx} = 4(\sec^{-1}(-5x + 3))^3(\frac{1}{|-5x + 3|\sqrt{(-5x + 3)^2 - 1}})(-5)$$

$$\frac{dy}{dx} = -\frac{20}{|-5x + 3|\sqrt{(-5x + 3)^2 - 1}} (\sec^{-1}(-5x + 3))^3$$

Teacher answers & working

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S3

Differentiate with respect to x.

$y = \sec^{-1}(7x + 4), \quad x = -1$

Answer:

$$y'(-1) = \frac{7}{|-3|\sqrt{(-3)^2 - 1}}$$

Working:

Let $u = 7x + 4$, so $u' = 7$.

Use the rule $\frac{d}{du}\sec^{-1}(u) = \frac{1}{|u|\sqrt{u^2 - 1}}$.

At $x = -1$, $u = 7 \cdot -1 + 4 = -3$.

$$\frac{dy}{du} = \frac{1}{|-3|\sqrt{(-3)^2 - 1}}$$

$$\frac{dy}{dx} = (\frac{1}{|-3|\sqrt{(-3)^2 - 1}})(7)$$

$$y'(-1) = \frac{7}{|-3|\sqrt{(-3)^2 - 1}}$$

S4

Differentiate with respect to x.

$y = \cot^{-1}(5x - 2)$

Answer:

$$\frac{dy}{dx} = -\frac{5}{1 + (5x - 2)^2}$$

Working:

Identify the inner function: $u = 5x - 2$.

Differentiate the inner function: $\frac{du}{dx} = 5$.

Select the inverse trig derivative rule: for $y = \cot^{-1}(u)$, $\frac{dy}{du} = -\frac{1}{1 + u^2}$.

Use the chain rule: $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$.

$$\frac{dy}{dx} = (-\frac{1}{1 + (5x - 2)^2})(5)$$

$$\frac{dy}{dx} = -\frac{5}{1 + (5x - 2)^2}$$

Teacher answers & working

Inverse Trig Differentiation · Core Derivatives and Evaluation

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E1

Differentiate with respect to x.

$y = (\cos^{-1}(9x+5))^4$

Answer:

$$\frac{dy}{dx} = -\frac{36}{\sqrt{1-(9x+5)^2}}(\cos^{-1}(9x+5))^3$$

Working:

This has an outer power and an inverse trig inner expression.

Let $v = \cos^{-1}(9x+5)$, so $y = v^4$ and $\frac{dy}{dv} = 4v^3$.

For $v = \cos^{-1}(u)$, $\frac{dv}{du} = -\frac{1}{\sqrt{1-u^2}}$.

The linear inner function is $u = 9x+5$, so $\frac{du}{dx} = 9$.

$$\frac{dy}{dx} = \frac{dy}{dv} \times \frac{dv}{du} \times \frac{du}{dx}$$

$$\frac{dy}{dx} = (4v^3)(-\frac{1}{\sqrt{1-u^2}})(9)$$

$$\frac{dy}{dx} = 4(\cos^{-1}(9x+5))^3(-\frac{1}{\sqrt{1-(9x+5)^2}})(9)$$

$$\frac{dy}{dx} = -\frac{36}{\sqrt{1-(9x+5)^2}}(\cos^{-1}(9x+5))^3$$

E2

Differentiate with respect to x.

$y = \sec^{-1}(5x-3), \quad x = 1$

Answer:

$$y'(1) = \frac{5}{|2|\sqrt{(2)^2-1}}$$

Working:

Let $u = 5x-3$, so $u' = 5$.

Use the rule $\frac{d}{du}\sec^{-1}(u) = \frac{1}{|u|\sqrt{u^2-1}}$.

At $x = 1$, $u = 5 \cdot 1 - 3 = 2$.

$$\frac{dy}{du} = \frac{1}{|2|\sqrt{(2)^2-1}}$$

$$\frac{dy}{dx} = (\frac{1}{|2|\sqrt{(2)^2-1}})(5)$$

$$y'(1) = \frac{5}{|2|\sqrt{(2)^2-1}}$$

Teacher answers & working

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E3

Differentiate with respect to x.

$y = \tan^{-1}(4x + 9)$

Answer:

$$\frac{dy}{dx} = \frac{4}{1 + (4x + 9)^2}$$

Working:

Identify the inner function: $u = 4x + 9$.

Differentiate the inner function: $\frac{du}{dx} = 4$.

Select the inverse trig derivative rule: for $y = \tan^{-1}(u)$, $\frac{dy}{du} = \frac{1}{1 + u^2}$.

Use the chain rule: $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$.

$$\frac{dy}{dx} = (\frac{1}{1 + (4x + 9)^2})(4)$$

$$\frac{dy}{dx} = \frac{4}{1 + (4x + 9)^2}$$